

SUMMARY

This video explains structured synthesis of advanced methodologies for constructing Minimal Deterministic Finite Automata (DFA) within Theory of Computation. The focus includes substring detection, parity constraints, compound prefix-length conditions, and nuanced minimization strategies.

Key insights:

The $m + n + 1$ rule determines states for combined prefix and modulo-length constraints. Languages equating counts of "01" and "10" are regular because their difference remains bounded.

Many textbook DFA diagrams are not minimal.

State efficiency requires structural reasoning, not memorization.

I. SUBSTRING CONSTRAINTS: DEBIT AND TRIBIT

Substring DFAs are built by constructing the minimal forward path and preserving partial progress upon mismatch.

I.1 Debit Substrings ("00" or "11")

Objective:

Accept strings containing either "00" or "11".

States Required: 4

Construction Logic:

Track single occurrences of 0 and 1.

If two identical consecutive symbols appear, transition to final state.

Final state has self-loop on both inputs.

I.2 Tribit Substrings ("000" or "111")

States Required: 6

Construction Logic:

Track first, second, and third consecutive identical symbols.

If interruption occurs, shift tracking rather than reset to start.

Final state self-loops once substring is detected.

II. EQUALITY OF SUBSTRING OCCURRENCES

Language:

Number of occurrences of “01” equals number of occurrences of “10”.

Regularity Insight:

The difference between counts of “01” and “10” in any binary string is always -1 , 0 , or 1 . Since this difference is bounded, the language is regular.

Accepted Strings:

ϵ
0
1
00
11
010
101

Rejected Strings:

01
10

Minimal DFA:

States Required: 5

Structure:

Initial state is final (accepts ϵ).

Separate branches for strings starting with 0 and 1.

Transitions balance the difference between substring counts.

III. PARITY AND MODULO CONSTRAINTS

Languages defined by parity of substring occurrences use cyclic structures.

Constraint	States	Key Logic
“10” occurs Even times	4	Accepts ϵ and zero occurrences
“10” occurs Odd times	4	Complement of Even DFA

Even and Odd DFAs share identical structure.
Final and non-final states are swapped for complement.

IV. COMPOUND CONSTRAINTS: START AND LENGTH DIVISIBILITY

Language type:

Strings start with substring S (length m)
Total length $\equiv r \pmod n$

IV.1 The $m + n + 1$ Rule

Total States = $m + n + 1$

Where:

m = prefix length
n = modulus
1 = dead state

IV.2 Application Examples

Starting Substring	Length Condition	Calculation	Total States
"a" (m = 1)	Divisible by 3 (n = 3)	$1 + 3 + 1$	5
"ba" (m = 2)	$2 \pmod 4$ (n = 4)	$2 + 4 + 1$	7
"aba" (m = 3)	$2 \pmod 5$ (n = 5)	$3 + 5 + 1$	9

Dead state ensures rejection for invalid prefixes.
Remaining states form cyclic length-tracking structure.

V. POSITIONAL "FOLLOWED BY" CONSTRAINTS

V.1 Every 'A' is Followed by 'B'

Minimal States: 3

States:

1. Final state (valid sequences and ϵ)
2. Non-final state after reading 'A'
3. Dead state if violation occurs

Transition Logic:

After 'A', must read 'B'.

If 'A' followed by 'A', transition to dead state.

V.2 Every 'A' is Never Followed by 'B'

States Required: 3

If pattern "AB" occurs → dead state.

V.3 Every 'A' is Followed by "BB"

States Required: 4

Additional state required to track first 'B' after 'A'.

VI. TECHNICAL CRITIQUE: GATE 2004 AMBIGUITY

Problem:

DFA intended for language:

Every 'A' is followed by at least two 'B's.

Issue:

If string contains no 'A', any number of 'B's should be accepted.

Exam diagram did not clearly mark all required final states.

Conclusion:

Diagram was incomplete.

Ambiguity likely resulted in universal marking.

VII. STRATEGIC INSIGHTS FOR MINIMIZATION

Principle 1: Build minimal path first.

Principle 2: Preserve partial progress instead of full reset.

Principle 3: Use complement logic for "Odd" vs "Even" conditions.

Principle 4: Apply $m + n + 1$ rule for compound prefix-length problems.

VIII. PRACTICE PROBLEMS

1. GATE 2017

Minimum states for $L = \{w_1 a w_2 \mid |w_1| = 2, |w_2| \geq 3\}$

2. GATE 2011

Minimum states for recognizing a^n where n is multiple of k ($k > 0$)

3. Construct DFA for strings that start with "AB" and end with "BA."

